# Md Nahid Husen Nayan

London, United Kingdom | nahidhusen23@gmail.com | linkedin.com/in/nahid-001-nayan

## PROFESSIONAL SUMMARY

First-Class Computer Science graduate and full-stack software engineer with hands-on experience building production-grade web applications and machine-learning systems. Skilled across the full stack (React, Next.js, Node.js, Spring Boot, Python) and the end-to-end ML lifecycle (data pipelines, model selection, evaluation, and SHAP explainability with scikit-learn, XGBoost, and LightGBM). Proven ability to own complex projects and deliver clear, reliable outcomes in cross-functional teams.

## EXPERIENCE

**Machine Learning Intern** Dec 2025 – Jan 2026

*Dalian University · Dalian, China*

- Took end-to-end ownership of a Python-based ML system for Type 2 diabetes risk prediction, covering problem definition, data pipeline design, feature engineering, model selection, evaluation, and documentation.
- Built robust data pipelines with pandas and NumPy to transform raw biomarker datasets into clean, analysis-ready inputs.
- Evaluated multiple classification algorithms via cross-validation to select the optimal model on measured evidence, working within a cross-functional international team across the full SDLC.

**Computer Lab Assistant** Oct 2024 – Apr 2025

*University of Roehampton · London, UK*

- Provided technical support to 100+ students across SQL, Linux, and Microsoft Azure, strengthening communication and structured problem-solving.
- Diagnosed and resolved front-end and software-environment issues methodically, maintaining consistency and reliability across lab systems.
- Identified recurring issues proactively and improved support documentation, demonstrating attention to detail and follow-through.

**Customer Assistant** May 2022 – Present

*Tesco · London, UK*

- Deliver reliable customer service in a fast-paced retail environment, demonstrating strong teamwork and time management alongside full-time study.

## EDUCATION

**BSc (Hons) Computer Science, First-Class Honours (1st)** — *University of Roehampton, London, UK* Sep 2022 – Jul 2025

- Relevant coursework: Algorithms & Data Structures, Software Engineering, Operating Systems, Databases, Web Development, Cybersecurity, Data Science, Machine Learning, and AI.
- Delivered production-grade full-stack apps in team and solo projects, from architecture through testing and deployment.

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript (ES6+), TypeScript, C/C++, SQL, HTML5, CSS3
**Frontend:** React.js, Next.js, Vue 3, Tailwind CSS, Bootstrap
**Backend:** Node.js, Express.js, Spring Boot, Python Flask, RESTful APIs, JWT Auth, Passport.js
**Databases & Search:** MongoDB, MySQL, SQL, Elasticsearch
**ML / AI:** scikit-learn, XGBoost, LightGBM, SHAP, NumPy, SciPy, TruncatedSVD, Collaborative Filtering
**Tools & DevOps:** Git, GitHub, Docker, Microsoft Azure, Jupyter Notebook, VS Code

## PROJECTS

**Book Search & Recommendation Engine**

*Full-stack discovery platform with ML-powered recommendations*

**Stack:** Next.js, React, Node.js, Express.js, MongoDB, Elasticsearch, Python, scikit-learn, Tailwind CSS, Zustand

- Indexed 240K+ books in Elasticsearch with multi-field boosted queries, ISBN fallback, and typo tolerance for sub-100ms search.
- Built an offline ML pipeline over 108M Goodreads interactions: SciPy sparse matrix → TruncatedSVD (k=50) → L2-normalised cosine similarity, writing top-20 similar books to MongoDB.
- Implemented JWT + Passport.js auth with wishlists, libraries, and custom shelves; interactions feed a reciprocal-rank aggregation engine for real-time personalised recommendations.

**Diabetes Disease Prediction System**

*Healthcare platform with dual-ML risk prediction and SHAP explainability*

**Stack:** Vue 3, Vite, Spring Boot, Java, Python Flask, XGBoost, LightGBM, SHAP, MySQL, MongoDB, ECharts

- Trained LightGBM and XGBoost models on lipid biomarkers (HDL, LDL, triglycerides) and used SHAP TreeExplainer to surface the top-10 contributing features per prediction for clinical transparency.
- Built a microservice architecture (Vue 3 + Spring Boot, independent Python Flask ML service) with ECharts dashboards and JWT multi-role RBAC across patient, doctor, and admin portals.